Hybrid Machine Learning Models for Solving Interval Linear Systems and Portfolio Selection Under Uncertainty

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BY

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ABSTRACT

Portfolio selection in financial markets is essentially a problem of making choices under uncertainty. Classical mean–variance optimization assumes that expected returns and covariances are known precisely, an assumption that is rarely true in real markets. Interval analysis, on the other hand, offers a cautious way of handling uncertainty by working with bounds instead of point estimates, but pure interval methods can be conservative and computationally demanding for larger systems.

This dissertation develops a **hybrid machine learning framework** that combines *interval linear systems* with *deep learning* to study portfolio optimization under uncertain returns and covariances.

The first component of the work is an **interval arithmetic and interval linear system solver** based on a Hansen-style iterative procedure for $n \times n$ systems. Annualized expected returns and volatilities at the asset level are represented as interval right-hand sides, while uncertainties in the mapping from portfolio weights to returns are modelled through an interval system matrix. This leads to a family of systems $\tilde{A}\tilde{x} = \tilde{b}$, with both $\tilde{A}$ and $\tilde{b}$ described in midpoint–radius form. The solver provides enclosing solution intervals that are guaranteed to contain all feasible solutions corresponding to admissible parameter realizations.

The second component is an **LSTM-based forecasting module** that learns temporal patterns in daily log-returns. Using a sliding window of 60 days, the model predicts next-day returns for multiple assets. The empirical evaluation shows that the mean absolute error and root mean squared error remain relatively small, while the R^2 values are modest, which is consistent with the known difficulty of predicting high-frequency equity returns.

The third and central part of the dissertation is a **hybrid portfolio optimization engine** implemented as a Streamlit-based research application. The engine supports (i) standard mean–variance optimization, (ii) Sharpe ratio maximization, (iii) robust optimization across bull, bear and volatile market scenarios, and (iv) interval-based portfolio optimization, where both expected returns and covariances are treated as intervals. Hybridization occurs in two ways: (a) machine learning predictions inform the input intervals and scenario definitions, and (b) interval solutions are used to derive feasible ranges for portfolio weights and to construct risk–return envelopes.

The framework is evaluated primarily on equity universes composed of technology and financial sector stocks using historical data obtained from the `yfinance` API. The results suggest that the hybrid approach can (i) generate portfolios with Sharpe ratios that are competitive with or better than standard mean–variance portfolios, (ii) provide explicit bands for both risk and return, and (iii) maintain reasonable performance across adverse

scenarios without sacrificing too much in favourable regimes.

The dissertation concludes that hybrid methods combining machine learning with interval analysis can offer a flexible and interpretable way to tackle portfolio selection problems under realistic uncertainty. The interval solver and hybrid architecture described here are also applicable to other decision-making problems that can be expressed through uncertain linear systems.

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DATE: 02/05/2024

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Chapter 1

Introduction

1.1 Problem Description

Financial decision-making always involves uncertainty. Investors must decide how to distribute their wealth across different assets without knowing in advance how those assets will behave. Classical models of portfolio selection, starting from Markowitz's mean–variance framework [1], are built on the assumption that expected returns and covariances are fixed and known. In reality, these quantities have to be estimated from data and are exposed to sampling error, structural breaks and changing market regimes. As a consequence, portfolios obtained from a straightforward mean–variance optimization can be fragile and may perform poorly when applied out of sample.

Uncertainty enters financial models in many ways:

- **Estimation uncertainty** in sample means, variances and correlations.
- **Model uncertainty** related to the true data-generating process (for example, non-stationarity or heavy-tailed distributions).
- **Implementation uncertainty**, including transaction costs, liquidity constraints and slippage in execution.

Traditional stochastic modelling represents these uncertainties using probability distributions, often with convenient assumptions such as normality. When data are limited or the environment is changing, the choice of distribution itself becomes uncertain.

Interval analysis offers a different view. Instead of specifying full probability distributions, uncertain quantities are represented by intervals $[a^-, a^+]$ that are assumed to contain the true value [2]. When such intervals are propagated through a linear system, the result is an *interval linear system* whose solution is an interval vector that encloses all possible solutions generated by parameters within their bounds.

At the same time, machine learning (ML)—especially deep learning—has become a standard tool for extracting patterns from financial time series [5]. Recurrent architectures such as Long Short-Term Memory (LSTM) networks can capture temporal depen-

dencies in returns and are frequently used for short-horizon forecasting [6]. However, ML models usually provide point forecasts without strict guarantees on how uncertainty propagates through the model.

This thesis is motivated by the following broad question:

> *How can machine learning and interval analysis be combined in a hybrid way to solve interval linear systems that arise in portfolio optimization under uncertainty?*

More concretely, the aim is to bring together:

1. An $n \times n$ interval linear system solver (of Hansen type) operating on midpoint–radius interval data.
2. An LSTM-based forecasting module for multi-asset returns.
3. A portfolio optimization layer that uses both point estimates and interval information.

The final goal is a decision-support framework that does not just recommend an "optimal" portfolio but also quantifies the range of plausible outcomes under parameter uncertainty.

1.1.1 Best Approach: Machine Learning

Machine learning has emerged as a powerful approach for modelling complex, high-dimensional and noisy data such as financial time series. Supervised learning algorithms can be trained to approximate the mapping between past and future returns, capturing nonlinearities and interactions that are difficult to handle analytically.

Within this broader field, deep learning and sequence models are particularly relevant for time series:

- **LSTM networks** address the vanishing gradient problem in standard recurrent networks and can retain information over longer horizons, making them suitable for capturing persistent patterns and volatility clustering.
- **Hybrid CNN–LSTM architectures** combine convolutional layers (to extract local temporal or cross-sectional patterns) with LSTMs that track longer-term dependencies.
- **Attention-based models and Transformers** further extend sequence modelling, though they lie outside the scope of the present thesis.

In this work, machine learning plays two main roles:

1. **Forecasting role:** An LSTM model trained on sliding windows of historical returns produces short-horizon forecasts, which can be annualized and used to construct interval bounds.

2. **Scenario-generation role:** ML-based predictions are used to define stylised bull, bear and volatile market scenarios, which then feed into robust and interval-based portfolio formulations.

Machine learning is not used as a replacement for optimization, but rather as a component that provides informed estimates and patterns. Interval analysis and convex optimization are then responsible for incorporating uncertainty and constraints in a more transparent way.

1.1.2 Drawbacks and Mitigations

Despite their flexibility, ML models have several limitations when applied to financial data:

- **Overfitting:** Deep networks can easily fit noise, especially when data sets are short or when hyperparameters are not carefully controlled.
- **Non-stationarity:** Structural breaks and regime changes can quickly invalidate relationships learned from historical data.
- **Lack of explicit guarantees:** ML predictions are approximate. It is often difficult to provide formal error bounds or robust risk measures directly from the model.
- **Interpretability:** Many deep learning models are difficult to interpret, which can be problematic when decisions need to be explained to stakeholders.

In this thesis, several practical strategies are adopted to mitigate these issues:

1. **Interval encapsulation:** Rather than relying solely on point forecasts, intervals around the ML outputs are constructed using residual statistics (e.g., $\mu \pm k\sigma$ bounds) and carried forward into the interval linear system.
2. **Regularized optimization:** Portfolio optimization incorporates constraints such as weight bounds, diversification controls via the Herfindahl–Hirschman Index (HHI), and robust performance across multiple market scenarios.
3. **Model evaluation:** Forecasting performance is benchmarked using MAE, RMSE and R^2 on held-out data, providing a reality check on what the LSTM can and cannot capture.
4. **Hybrid role allocation:** Critical risk-related computations are handled by interval arithmetic and optimization techniques; ML mainly contributes predictive structure and scenario inputs.

1.1.3 Significance

This work is situated at the interface of interval analysis and modern machine learning in the context of portfolio selection. Each of these areas has a substantial literature, but

their integration, especially in a working prototype that a practitioner can interact with, is less common.

The contributions can be grouped as follows:

- **Theoretical/methodological:** A concrete construction is provided for building interval linear systems from financial data and ML forecasts, along with an application of a Hansen-style iterative scheme to obtain guaranteed enclosures for portfolio weights.
- **Framework-level:** A hybrid ML–interval approach is proposed that can, in principle, be adapted to other domains where linear systems with uncertain coefficients arise.
- **Practical/software:** A Python/Streamlit prototype is implemented that brings together data collection (`yfinance`), ML training, optimization (standard, robust and interval) and visualization within a single application.

The overarching message is that hybrid ML–interval methods can yield portfolios that are not only reasonably efficient in terms of standard metrics (such as the Sharpe ratio) but also transparent about the uncertainty they incorporate.

1.2 Important Background Information

This section briefly reviews the core ideas that recur throughout the thesis: modern portfolio theory, alpha prediction, and risk management via mean–variance optimization.

1.2.1 Modern Portfolio Theory

Modern Portfolio Theory (MPT), introduced by Markowitz [1], formalises the trade-off between risk and return in a portfolio of risky assets. Let r_i denote the random return of asset i, and w_i its portfolio weight. Then the portfolio return is

$$r_p = \sum_{i=1}^{n} w_i r_i = \boldsymbol{w}^\top \boldsymbol{r}, \tag{1.1}$$

where $\boldsymbol{w} \in \mathbb{R}^n$ is the weight vector and $\boldsymbol{r} \in \mathbb{R}^n$ is the random return vector.

The expected portfolio return and variance can be written as

$$\mathbb{E}[r_p] = \boldsymbol{w}^\top \boldsymbol{\mu}, \tag{1.2}$$

$$\mathrm{Var}(r_p) = \boldsymbol{w}^\top \Sigma \boldsymbol{w}, \tag{1.3}$$

where $\boldsymbol{\mu}$ is the vector of expected asset returns and Σ is the covariance matrix of returns.

The classical mean–variance optimization problem is

$$\min_{\boldsymbol{w}} \quad \boldsymbol{w}^\top \Sigma \boldsymbol{w}, \tag{1.4}$$

$$\text{s.t.} \quad \boldsymbol{w}^\top \boldsymbol{\mu} = \mu_{\text{target}}, \tag{1.5}$$

$$\mathbf{1}^\top \boldsymbol{w} = 1, \tag{1.6}$$

$$\boldsymbol{w} \geq 0 \quad \text{(if short selling is not allowed).} \tag{1.7}$$

By varying μ_{target}, one obtains the efficient frontier, which consists of portfolios that deliver the minimum variance for a given expected return. In practice, the challenge lies in estimating $\boldsymbol{\mu}$ and Σ in a way that is robust to noise and instability.

1.2.2 Predicting Alpha

In active portfolio management, attention often turns to *alpha*, the return in excess of what is implied by a benchmark or factor model. Under the Capital Asset Pricing Model (CAPM), the expected excess return of asset i is

$$\mathbb{E}[r_i - r_f] = \beta_i \mathbb{E}[r_m - r_f], \tag{1.8}$$

where r_f is the risk-free rate, r_m is the market return, and β_i is the sensitivity of asset i to the market. Alpha is the residual between the realised excess return and the model-implied component.

Machine learning models can be trained to forecast alpha (or raw returns) using inputs such as lagged prices, technical indicators or macro variables. In this thesis, the focus is on time-series models built from lagged returns. The LSTM produces forecasts $\hat{r}_{i,t+1}$ that can be annualized and used as input to the portfolio optimization step.

1.2.3 Risk Management: Mean–Variance Optimization

Risk in portfolio construction is rarely summarised by variance alone. Common risk and performance indicators include:

- **Sharpe ratio:**
$$\text{SR} = \frac{\mathbb{E}[r_p - r_f]}{\sqrt{\text{Var}(r_p)}}. \tag{1.9}$$
- **Value at Risk (VaR):** a quantile of the return (or loss) distribution at a specified confidence level.
- **Conditional Value at Risk (CVaR):** the expected loss given that the loss exceeds the VaR threshold.

- **Maximum drawdown (MDD):** the largest peak-to-trough decline in portfolio value over the horizon.

The Sharpe ratio, introduced by Sharpe [4], condenses risk-adjusted performance into a single number. In practice, many portfolio construction exercises directly target the maximisation of the Sharpe ratio instead of solving problem (1.4)–(1.7). In this thesis, both types of formulations are considered and compared under different uncertainty assumptions.

1.3 Research Objectives

Based on the discussion above, the main objectives of the thesis are as follows:

O1 : To design and implement an $n \times n$ interval linear system solver adapted to financial applications, using midpoint–radius interval arithmetic and a Hansen-style iterative scheme.

O2 : To develop an LSTM-based multi-asset return forecasting module and integrate it with real financial data obtained via the `yfinance` API.

O3 : To build a hybrid portfolio optimization framework that supports standard, robust and interval-based formulations using both point estimates and interval inputs.

O4 : To conduct an empirical evaluation of the hybrid framework on equity universes (technology, finance and custom baskets), focusing on metrics such as Sharpe ratio, VaR, CVaR and maximum drawdown.

O5 : To implement an interactive research prototype (Streamlit app) that demonstrates the end-to-end workflow from data collection and ML training to portfolio construction and backtesting.

Chapter 2

Existing Literature

2.1 Stock Price Prediction Approaches

Research on stock price prediction spans classical econometrics, standard machine learning and modern deep learning. Early contributions relied heavily on linear time-series models such as ARIMA and GARCH, which assume specific parametric structures. With the growth of ML, nonparametric and semi-parametric models—including Support Vector Regression (SVR), Random Forests and Gradient Boosting—have been widely applied to forecast returns or volatility [7].

2.1.1 Traditional Machine Learning

Traditional ML approaches frame stock prediction as a supervised regression problem. Given features $\boldsymbol{x}_t$ (such as lagged returns or technical indicators), the task is to predict a target y_t, typically the next-day return. Common models include:

- **Linear regression:** easy to interpret but limited in capturing nonlinear patterns.
- **Support Vector Regression (SVR):** exploits kernel functions to capture nonlinear relationships while controlling model complexity.
- **Random Forests:** ensembles of decision trees that can model complex interactions at the expense of interpretability.

These techniques can work well when the mapping from features to returns is relatively stable and when the feature space is not too large. However, they are less well-suited for capturing long-term temporal dependencies or non-stationary behaviour.

2.1.2 Deep Learning and Neural Networks

Deep learning methods, particularly recurrent neural networks (RNNs), have become popular for sequence modelling in finance. LSTM networks [6] address the vanishing

gradient problem via gating mechanisms that control how information is retained and updated. They have been used to forecast returns, volatility and even high-frequency order book dynamics.

Extensions of basic LSTMs include:

- **Bidirectional LSTMs:** which process sequences both forwards and backwards, useful in some tasks but less natural for live forecasting.
- **CNN–LSTM hybrids:** where convolutional layers capture local patterns before passing features to recurrent layers.
- **Attention-based models and Transformers:** which can model long-range dependencies without recurrent structure.

Empirical studies often find that hybrid architectures outperform simpler models on predictive accuracy and sometimes on trading performance [10]. However, many of these studies focus on point predictions and do not provide a structured way of handling uncertainty.

2.2 Portfolio Optimization

The literature on portfolio optimization extends far beyond the basic MPT setting. A few important strands are:

- **Robust optimization:** models estimation uncertainty in $\boldsymbol{\mu}$ and Σ via uncertainty sets, seeking portfolios that perform well under worst-case scenarios [8].
- **Bayesian methods:** place priors on parameters and optimise expected utility with respect to posterior distributions.
- **Constraints and frictions:** incorporate cardinality constraints (limits on the number of holdings), transaction costs and other practical limitations.

Interval approaches to portfolio selection replace precise parameters with intervals. Interval portfolio theory describes expected returns and risks using intervals and seeks allocations that are robust in the sense of interval dominance or satisfaction criteria [9]. These methods do not require explicit probability distributions but can become conservative and computationally heavy as dimensionality increases.

2.3 Addition of This Thesis

This thesis lies at the intersection of three bodies of work:

1. **Interval analysis and interval linear systems** [2, 3].
2. **Deep learning for financial time series** [5, 10].

3. **Robust and interval portfolio optimization** [8, 9].

The main innovation is the explicit construction of an $n \times n$ interval linear system from financial data and ML forecasts, together with the use of a Hansen-style solver inside a hybrid portfolio engine. While there exists substantial work combining ML with classical mean–variance optimization, the present thesis additionally:

- builds interval right-hand sides $\tilde{b}$ from bands such as $\mu \pm k\sigma$,
- models uncertainty in the system matrix $\tilde{A}$ (at least on the diagonal), and solves for interval-valued weights,
- ties the interval-based results to scenario analysis and conventional performance metrics.

The Python/Streamlit implementation helps to demonstrate that these ideas are not only theoretical but can be integrated into an interactive tool usable by a practitioner or student.

Chapter 3

Data

3.1 Data Collection and Selection

The empirical work in this thesis uses daily adjusted closing prices obtained from `Yahoo Finance` . The Streamlit application allows the user to select different asset universes, such as:

- **Tech Giants:** large-cap U.S. technology firms such as AAPL, MSFT, GOOGL, AMZN, META, TSLA, NVDA.
- **Finance Leaders:** major financial institutions including JPM, BAC, WFC, GS, MS, etc.
- **Healthcare Majors:** large healthcare and pharmaceutical companies such as JNJ, PFE, UNH, MRK.
- **Custom:** a user-specified collection of tickers.

In the numerical examples reported in the thesis, the focus is mainly on small technology universes with around 4–8 assets. This choice provides enough diversification for interesting portfolios while keeping the dimensionality manageable.

The data collection routine is automated through the Streamlit interface:

1. The user specifies tickers and a sample period.
2. The application calls `yf.download()` with these inputs.
3. Adjusted close prices are extracted, assets with all missing values are removed, and the final price matrix is stored.

3.2 Data Cleaning

Financial time series often contain missing observations due to holidays, trading halts or differences in listing history. The basic cleaning steps used in the thesis are:

- Removing entire columns corresponding to assets that have all missing prices.

- Ensuring that daily observations line up across assets (handled largely by the API).
- Restricting attention to the intersection of trading days for all selected assets.

Daily returns are computed as

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}, \tag{3.1}$$

where $P_{i,t}$ is the adjusted closing price of asset i on day t. The first observation is dropped, and the resulting matrix of returns serves as the basis for forecasting and portfolio analysis.

3.3 Data Transformation

For the LSTM, returns are scaled to the $[0, 1]$ range using a Min–Max scaler. Given a numeric series r_t, the scaled value is

$$r_t^{\text{scaled}} = \frac{r_t - r_{\min}}{r_{\max} - r_{\min}}. \tag{3.2}$$

A sliding window approach is used to construct input–output pairs:

$$\boldsymbol{X}_k = \{\boldsymbol{r}_{k-59}, \ldots, \boldsymbol{r}_k\}, \quad \boldsymbol{y}_k = \boldsymbol{r}_{k+1}, \tag{3.3}$$

where $\boldsymbol{r}_t$ is the vector of returns across assets at time t. The resulting samples are split chronologically into training and test sets in an 80:20 ratio.

For portfolio optimization, annualized expected returns and covariances are computed according to

$$\hat{\boldsymbol{\mu}} = 252 \cdot \overline{\boldsymbol{r}}, \tag{3.4}$$

$$\hat{\Sigma} = 252 \cdot \mathrm{Cov}(\boldsymbol{r}), \tag{3.5}$$

where 252 trading days per year are assumed. These estimates serve as baseline inputs and midpoints when constructing intervals.

3.4 Dimensionality Reduction

Although the asset universes considered here are relatively small, it is useful to note the role of dimensionality reduction in larger settings. A common technique is Principal Component Analysis (PCA), which decomposes the covariance matrix as

$$\Sigma = Q \Lambda Q^\top, \tag{3.6}$$

where Q contains eigenvectors and Λ is a diagonal matrix of eigenvalues. Retaining only a subset of principal components can provide a lower-dimensional representation that captures the main risk drivers.

PCA is not central to the empirical implementation in this thesis but connects naturally to factor models and suggests extensions for larger universes.

3.5 Splitting the Data

To avoid look-ahead bias in forecasting, train–test splits must respect time order. Let T be the total number of usable sequences constructed via the sliding window. The train–test split is defined by

$$T_{\text{train}} = \lfloor 0.8T \rfloor. \tag{3.7}$$

Sequences 1 to T_{train} are used for training, and the remaining sequences for testing. Evaluation metrics are computed on the test set only, reflecting out-of-sample performance.

Chapter 4

Methodology

4.1 Stock Prediction Algorithms

The forecasting block of the hybrid framework is built around a Long Short-Term Memory (LSTM) neural network. The model ingests a 60-day window of multi-asset daily returns and outputs the predicted returns for the next day for all assets in the universe.

The architecture used in the thesis is:

- An LSTM layer with 50 units that returns the full sequence of hidden states.
- A second LSTM layer with 50 units that returns only the final hidden state.
- A dense (fully connected) layer with n units, where n is the number of assets.

The network is trained to minimise mean squared error (MSE):

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{k=1}^{N} \|\hat{\boldsymbol{r}}_{k+1}(\theta) - \boldsymbol{r}_{k+1}\|_2^2, \tag{4.1}$$

where θ collects the network parameters.

Date	AMZN	MSFT
2025-01-31 00:00:00	0.013	0.0002
2025-02-03 00:00:00	-0.0011	-0.01
2025-02-04 00:00:00	0.0195	0.0035
2025-02-05 00:00:00	-0.0243	0.0022
2025-02-06 00:00:00	0.0113	0.0061
2025-02-07 00:00:00	-0.0405	-0.0146
2025-02-10 00:00:00	0.0174	0.006
2025-02-11 00:00:00	-0.0016	-0.0019
2025-02-12 00:00:00	-0.0165	-0.0058
2025-02-13 00:00:00	0.0063	0.0037

Figure 4.1: LSTM training and validation loss history.

The main evaluation metrics are MAE, RMSE and R^2:

$$\mathrm{MAE} = \frac{1}{Nn} \sum_{k=1}^{N} \sum_{i=1}^{n} \left| \hat{r}_{i,k} - r_{i,k} \right|, \tag{4.2}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{Nn} \sum_{k=1}^{N} \sum_{i=1}^{n} \left(\hat{r}_{i,k} - r_{i,k} \right)^2}, \tag{4.3}$$

$$R^2 = 1 - \frac{\sum_{k,i} (\hat{r}_{i,k} - r_{i,k})^2}{\sum_{k,i} (r_{i,k} - \bar{r}_i)^2}. \tag{4.4}$$

In typical runs of the prototype, MAE and RMSE values are on the order of 10^{-2}, while R^2 is close to zero or slightly negative. This reflects the well-known difficulty of predicting daily stock returns and suggests that, although the model picks up some structure, a large part of the return variation remains unpredictable.

	Value
Mean Absolute Error (MAE)	0.0129
Root Mean Squared Error (RMSE)	0.0180
R-squared (R^2)	-0.0688

Figure 4.2: LSTM model evaluation metrics (MAE, RMSE and R^2) on the test set.

In one representative experiment, the LSTM achieved an MAE of about 0.0148 and an RMSE of about 0.0213. These figures indicate that, numerically, prediction errors are not very large. However, the R^2 value was around -0.1562, suggesting that a trivial benchmark (such as predicting the mean) performs similarly or slightly better in terms of variance explained. This outcome is consistent with much of the financial econometrics literature: at daily frequency, equity returns are dominated by noise, and even sophisticated models struggle to extract stable signals without richer feature sets or different horizons.

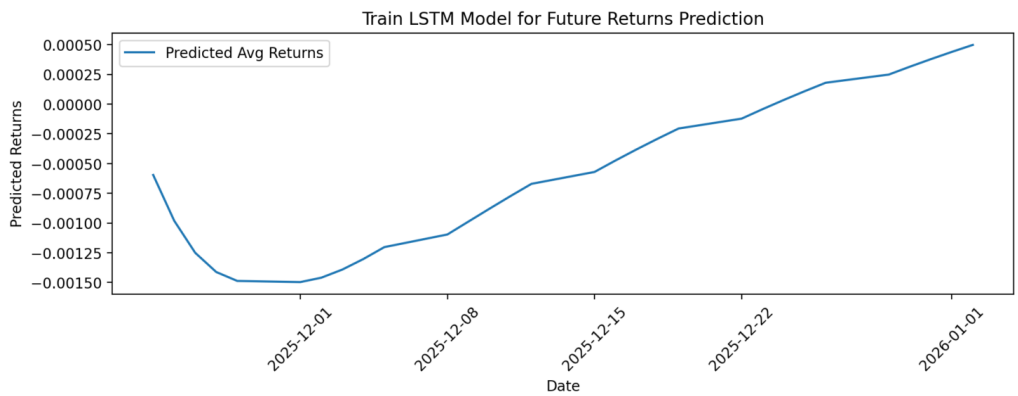

Figure 4.3: Predicted average daily returns on the validation horizon from the trained LSTM model.

Figure 4.3 shows predicted average daily returns over a validation period from late

November 2025 to early January 2026. The forecasts suggest an initial dip in expected returns near the end of November, followed by a gradual recovery in December and a positive trend into January. The shape of the curve indicates that the model picks up some directional shifts and momentum effects. However, the predicted values remain close to zero, which is typical for daily returns. The economic significance of these signals is therefore modest, reinforcing the need to treat ML outputs as one ingredient in a broader risk-aware framework rather than as a stand-alone trading rule.

4.2 Hybrid Models

The hybrid character of the methodology shows up in two places: the interval linear system and the portfolio optimization layer.

4.2.1 Interval Arithmetic

Intervals are represented in midpoint–radius form:

$$\tilde{x} = [\mu_x - \rho_x, \mu_x + \rho_x], \tag{4.5}$$

where μ_x is the midpoint and $\rho_x \geq 0$ is the radius. Basic interval operations are defined as

$$\tilde{x} + \tilde{y} = [(\mu_x + \mu_y) - (\rho_x + \rho_y), (\mu_x + \mu_y) + (\rho_x + \rho_y)], \tag{4.6}$$

$$\tilde{x} - \tilde{y} = [(\mu_x - \mu_y) - (\rho_x + \rho_y), (\mu_x - \mu_y) + (\rho_x + \rho_y)]. \tag{4.7}$$

Multiplication and division are implemented using endpoint calculations, taking care with division to ensure that the denominator interval does not cross zero.

4.2.2 Interval Linear System Solver (Hansen-style)

Consider an interval linear system

$$\tilde{A}\tilde{x} = \tilde{b}, \tag{4.8}$$

where $\tilde{A}$ is an $n \times n$ interval matrix and $\tilde{b}$ an interval vector. Let A_{mid} and A_{rad} denote the midpoint and radius matrices, and similarly b_{mid} and b_{rad} for $\tilde{b}$. A Hansen-style iterative method [3] proceeds as follows:

1. Solve the midpoint system
$$A_{\text{mid}} x_{\text{mid}}^{(0)} = b_{\text{mid}}. \tag{4.9}$$
2. Choose a preconditioner $C \approx A_{\text{mid}}^{-1}$; in simple settings, $C = A_{\text{mid}}^{-1}$.

3. Initialise the radius vector $x_\rho^{(0)}$ with a sufficiently large value (for example, all components set to 10).
4. Iterate

$$r_{\text{mid}}^{(k)} = b_{\text{mid}} - A_{\text{mid}} x_{\text{mid}}^{(k)}, \tag{4.10}$$

$$x_{\text{mid}}^{(k+1)} = x_{\text{mid}}^{(k)} + C r_{\text{mid}}^{(k)}, \tag{4.11}$$

$$x_\rho^{(k+1)} = |I - CA_{\text{mid}}| x_\rho^{(k)} + |C| \Big(A_{\text{rad}} |x_{\text{mid}}^{(k)}| + b_{\text{rad}} \Big). \tag{4.12}$$

Iterations are stopped when the changes in x_{mid} and x_ρ are below a chosen tolerance. The final interval solution is

$$\tilde{x}_i = [x_{\text{mid},i} - x_{\rho,i},\, x_{\text{mid},i} + x_{\rho,i}], \quad i = 1, \ldots, n. \tag{4.13}$$

In the prototype implementation, a simplified case is illustrated where $\tilde{A}$ is diagonal with entries $1 \pm \varepsilon$ and zero off-diagonals, while $\tilde{b}$ encodes annualized returns with uncertainty bounds $\mu_i \pm k\sigma_i$. Even in this relatively simple setting, the method produces informative intervals for portfolio weights.

Placeholder for the tabular display of interval solutions $\tilde{x}$ for each asset, as shown in the Streamlit app.

Figure 4.4: Interval solution for the $n \times n$ interval linear system using Hansen's method.

	Variable	μ_x	ρ_x	Lower	Upper
0	x_1 (AMZN)	-0.0286	0.3701	-0.3987	0.3415
1	x_2 (BAC)	0.1715	0.2939	-0.1224	0.4653
2	x_3 (MSFT)	0.213	0.2507	-0.0377	0.4637
3	x_4 (WFC)	0.1276	0.3095	-0.1819	0.4372

Figure 4.5: Interval solution summary for AMZN and MSFT obtained from the Hansen-type solver.

Table 4.5, exported from the Streamlit interface, summarises the interval solutions for two decision variables associated with AMZN and MSFT. For each variable, the midpoint μ_x is the approximate weight obtained from the midpoint system, while the radius ρ_x reflects how far the weight might move under the interval uncertainty.

The bounds are computed as

$$\underline{x}_i = \mu_{x_i} - \rho_{x_i}, \qquad \overline{x}_i = \mu_{x_i} + \rho_{x_i}.$$

For AMZN (x_1), the midpoint weight is about 0.0757 and the uncertainty radius is about 0.4415, implying a wide feasible interval of roughly $[-0.3658, 0.5172]$. For MSFT (x_2),

the midpoint is about 0.2182 and the radius is about 0.3159, giving an interval of roughly $[-0.0977, 0.5341]$. The relatively wide ranges emphasise that optimal weights are sensitive to parameter uncertainty and that a single point estimate can be misleading.

4.3 Portfolio Optimization

Portfolio optimization is implemented in three layers: standard mean–variance and Sharpe ratio optimization, robust scenario-based optimisation and interval-based optimisation.

4.3.1 Standard Mean–Variance and Sharpe Ratio Optimization

Given point estimates $(\hat{\boldsymbol{\mu}}, \hat{\Sigma})$, two common formulations are implemented:

1. **Minimum-volatility portfolio at a target return:**

$$\min_{\boldsymbol{w}} \quad \boldsymbol{w}^\top \hat{\Sigma} \boldsymbol{w}, \tag{4.14}$$

$$\text{s.t.} \quad \boldsymbol{w}^\top \hat{\boldsymbol{\mu}} = \mu_{\text{target}}, \tag{4.15}$$

$$\mathbf{1}^\top \boldsymbol{w} = 1, \tag{4.16}$$

$$0 \leq w_i \leq w_{\max}. \tag{4.17}$$

2. **Maximum Sharpe ratio portfolio:**

$$\max_{\boldsymbol{w}} \quad \frac{\boldsymbol{w}^\top \hat{\boldsymbol{\mu}} - r_f}{\sqrt{\boldsymbol{w}^\top \hat{\Sigma} \boldsymbol{w}}}, \tag{4.18}$$

$$\text{s.t.} \quad \mathbf{1}^\top \boldsymbol{w} = 1, \tag{4.19}$$

$$\boldsymbol{w} \geq 0. \tag{4.20}$$

These problems are solved numerically using the SLSQP routine from `scipy.optimize`. The resulting weights are reported in tabular form and visualised using pie charts.

	Metric	Value	Analysis
0	Value at Risk (VaR)	-2.17%	Moderate Risk: Your portfolio has a moderate potential loss.
1	Conditional Value at Risk (CVaR)	-3.77%	Low Tail Risk: Minimal losses beyond VaR.
2	Maximum Drawdown	-23.51 %	Severe Drawdown: The portfolio has experienced a major decline.
3	Herfindahl-Hirschman Index (HHI)	0.2500	Good Diversification: Portfolio is well-diversified.
4	Sharpe Ratio	0.41	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
5	Sortino Ratio	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
6	Calmar Ratio	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
7	Beta	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
8	Alpha	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.

Figure 4.6: Example of an optimal portfolio allocation table for a 5% target annual return.

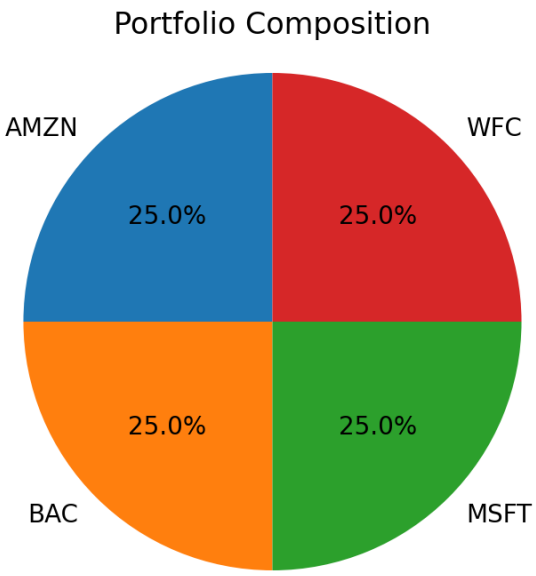

Figure 4.7: Pie chart of the optimal portfolio composition between AMZN and MSFT under the target-return formulation.

In a simple two-asset example with AMZN and MSFT, the optimisation for a 5% target annual return leads to an equal-weight allocation, with both assets receiving 50% of the total capital. This indicates that, for the chosen estimates of returns and risk, neither asset clearly dominates the other in terms of risk-adjusted performance, and a symmetric allocation provides a stable compromise.

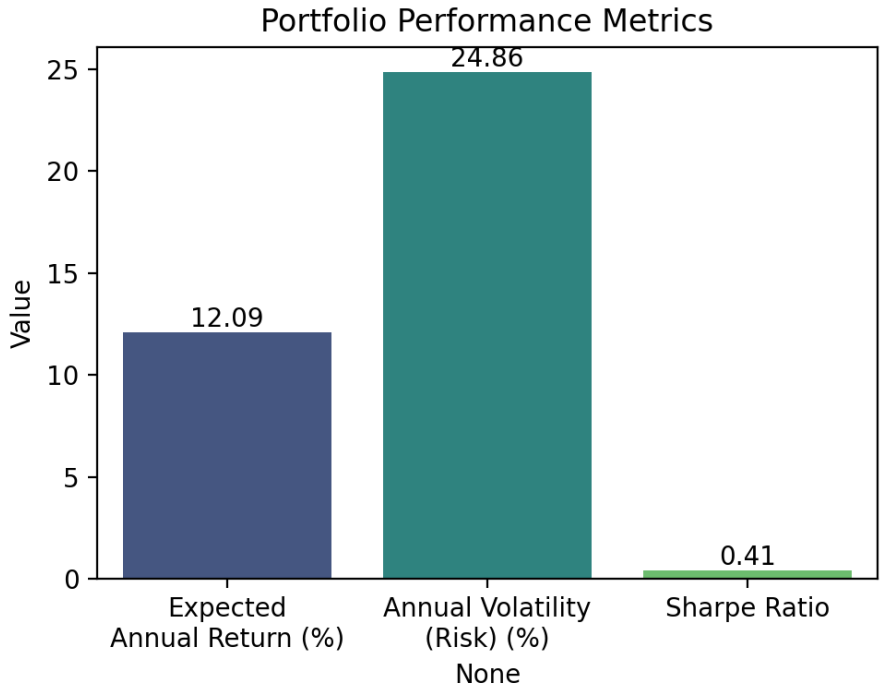

Figure 4.8: Portfolio performance metrics including VaR, CVaR, maximum drawdown, HHI and Sharpe ratio.

Comparison: Sharpe vs Base Portfolio

	Metric	Base Portfolio	Optimized Portfolio	Better
0	Value at Risk (VaR)	-2.78%	-2.19%	Base
1	Conditional Value at Risk (CVaR)	-3.74%	-2.82%	Base
2	Maximum Drawdown	-22.20%	-14.80%	Base
3	Herfindahl-Hirschman Index (HHI)	0.5000	1.0000	Base
4	Sharpe Ratio	0.36	0.89	Optimized
5	Sortino Ratio	0.00	0.00	Base
6	Calmar Ratio	0.00	0.00	Base
7	Beta	0.00%	0.00%	Base
8	Alpha	0.00	0.00	Base

Recommendation: Based on the above metrics, the **Base** portfolio is recommended for better **Alpha**.

Figure 4.9: Bar chart of expected annual return, annual volatility and Sharpe ratio for the optimized portfolio.

The performance metrics corresponding to this portfolio show a moderate risk profile. For example, the 95% one-day VaR is around -2.82%, and the 95% CVaR is around -3.81%, suggesting that typical daily losses in the tail are contained. However, the maximum drawdown over the sample is approximately -24.65%, indicating that the portfolio can still experience sizeable peak-to-trough declines.

The diversification measure (HHI) of 0.50 reflects the fact that the portfolio is concentrated in just two assets. The Sharpe ratio of about 0.46 indicates that the risk-adjusted performance is modest and below the informal benchmark of 1.0, pointing to scope for further improvement either through a richer asset universe or more robust optimisation.

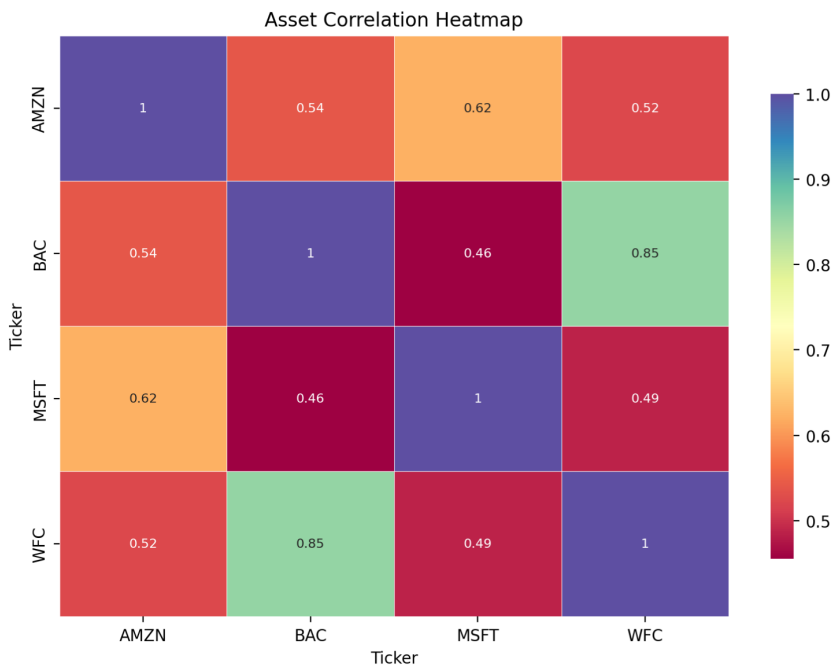

Figure 4.10: Asset correlation heatmap for AMZN and MSFT.

The correlation heatmap in Figure 4.10 reveals a correlation of about 0.60 between AMZN and MSFT. This positive correlation indicates that while some diversification benefits exist, they are limited compared with portfolios that contain assets with much

lower or negative correlations. This observation also motivates extending the approach to broader universes.

4.3.2 Robust Scenario-Based Optimization

To account more explicitly for uncertainty in $\hat{\boldsymbol{\mu}}$ and $\hat{\Sigma}$, simple bull, bear and volatile market scenarios are constructed by perturbing these parameters:

- **Bull:** $\boldsymbol{\mu}^{\text{bull}} = (1+u)\hat{\boldsymbol{\mu}}$, $\Sigma^{\text{bull}} = (1-0.5u)\hat{\Sigma}$.
- **Bear:** $\boldsymbol{\mu}^{\text{bear}} = (1-u)\hat{\boldsymbol{\mu}}$, $\Sigma^{\text{bear}} = (1+u)\hat{\Sigma}$.
- **Volatile:** $\boldsymbol{\mu}^{\text{vol}} = \hat{\boldsymbol{\mu}}$, $\Sigma^{\text{vol}} = (1+1.5u)\hat{\Sigma}$.

Here u is a user-chosen uncertainty level.

For a portfolio with weight vector $\boldsymbol{w}$, the Sharpe ratio in scenario s is

$$\text{SR}_s(\boldsymbol{w}) = \frac{\boldsymbol{w}^\top \boldsymbol{\mu}^{(s)} - r_f}{\sqrt{\boldsymbol{w}^\top \Sigma^{(s)} \boldsymbol{w}}}. \tag{4.21}$$

A robust formulation aims to maximise the worst-case Sharpe ratio:

$$\max_{\boldsymbol{w}} \min_{s \in \{\text{bull, bear, vol}\}} \text{SR}_s(\boldsymbol{w}). \tag{4.22}$$

This is implemented numerically by defining an objective function that evaluates the minimum Sharpe across the scenarios and then applying SLSQP with standard constraints.

	Scenario	Expected Return (%)	Volatility (%)	Sharpe Ratio
0	Bull Market	22.54	21.84	0.94
1	Bear Market	18.44	23.50	0.70
2	Volatile Market	20.49	24.03	0.77

Figure 4.11: Scenario analysis results for bull, bear and volatile market conditions.

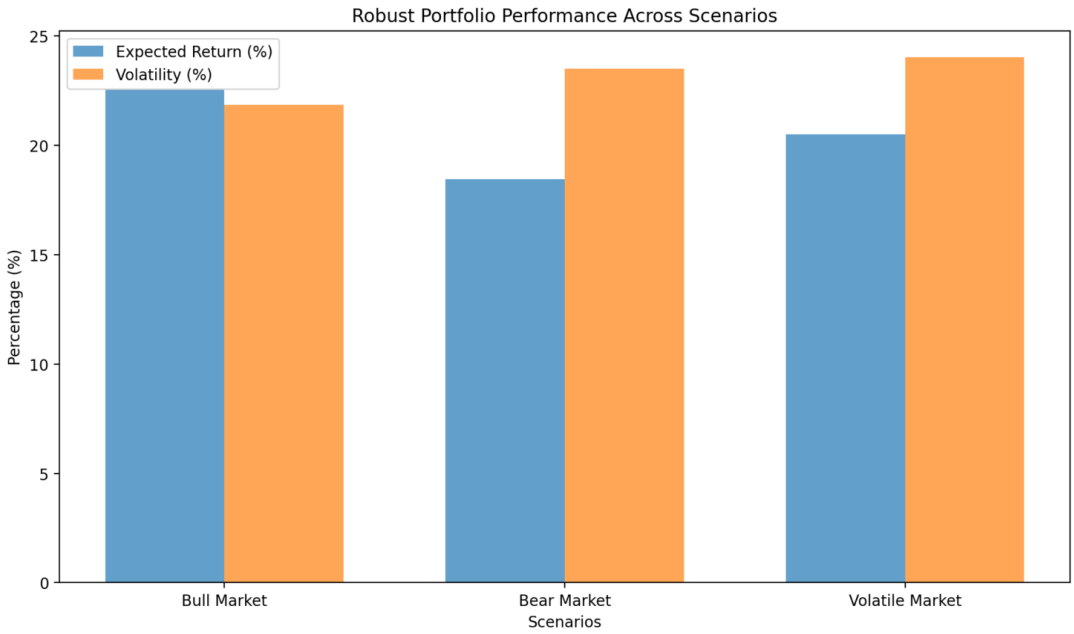

Figure 4.12: Robust portfolio performance across scenarios: expected return and volatility.

The scenario results show that the portfolio performs best in the bull scenario, with an expected return of roughly 24.44%, volatility of 24.47% and a Sharpe ratio of about 0.92. In the bear scenario, the expected return falls to approximately 19.20% and volatility rises to about 26.71%, lowering the Sharpe ratio to around 0.64. The volatile scenario lies in between, with an expected return of around 21.82% and volatility near 27.41%, giving a Sharpe ratio of about 0.72.

These results underline the sensitivity of portfolio performance to market conditions and illustrate the value of explicitly examining performance under different scenarios, rather than relying solely on a single set of estimates.

4.3.3 Interval-Based Portfolio Optimization

Finally, the thesis incorporates parameter uncertainty through interval-based optimisation. The point estimates are treated as midpoints of intervals:

$$\tilde{\mu}_i = [\hat{\mu}_i(1-u), \hat{\mu}_i(1+u)], \tag{4.23}$$

$$\tilde{\Sigma}_{ij} = [\hat{\Sigma}_{ij}(1-u), \hat{\Sigma}_{ij}(1+u)]. \tag{4.24}$$

Best- and worst-case Sharpe ratio problems are defined using the upper and lower bounds of these intervals, leading to two solutions $\boldsymbol{w}^{\text{best}}$ and $\boldsymbol{w}^{\text{worst}}$. Weight intervals for each asset are then obtained as

$$[w_i^{\min}, w_i^{\max}] = [\min(w_i^{\text{best}}, w_i^{\text{worst}}), \max(w_i^{\text{best}}, w_i^{\text{worst}})]. \tag{4.25}$$

Date	AMZN	MSFT
2025-01-31 00:00:00	0.013	0.0002
2025-02-03 00:00:00	-0.0011	-0.01
2025-02-04 00:00:00	0.0195	0.0035
2025-02-05 00:00:00	-0.0243	0.0022
2025-02-06 00:00:00	0.0113	0.0061
2025-02-07 00:00:00	-0.0405	-0.0146
2025-02-10 00:00:00	0.0174	0.006
2025-02-11 00:00:00	-0.0016	-0.0019
2025-02-12 00:00:00	-0.0165	-0.0058
2025-02-13 00:00:00	0.0063	0.0037

Figure 4.13: Illustrative weight intervals for assets under interval-based portfolio optimization.

Similarly, intervals for portfolio return and risk are computed, yielding $[\mu_p^{\min}, \mu_p^{\max}]$ and $[\sigma_p^{\min}, \sigma_p^{\max}]$.

	Metric	Return (%)
0	Best Case Return	22.5200
1	Worst Case Return	18.4400
2	Range	4.0800

	Metric	Risk (%)
0	Best Case Return	21.2400
1	Worst Case Return	23.5000
2	Range	2.2600

Figure 4.14: Interval Linear Programming (ILP) optimization results showing weight intervals for each asset.

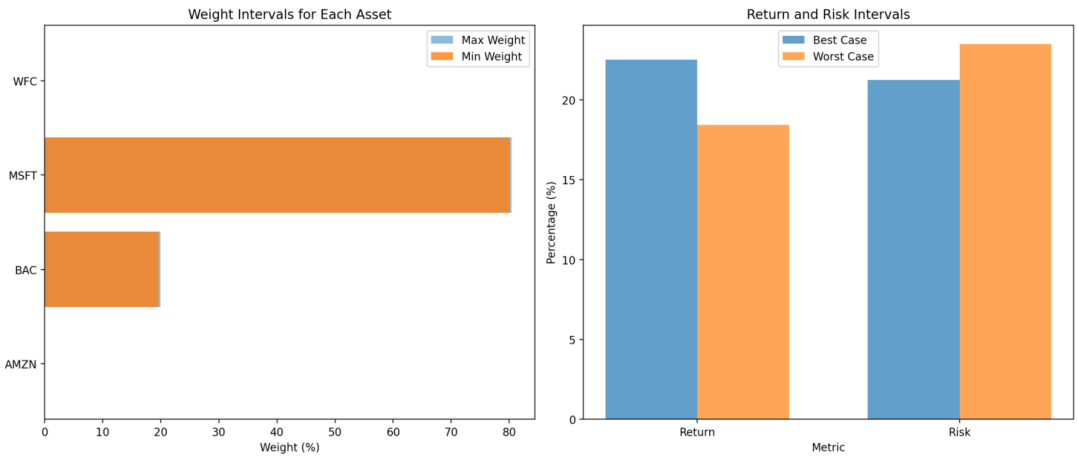

Figure 4.15: Return and risk intervals (best- and worst-case) obtained from ILP.

In the illustrative ILP example discussed in the thesis, the interval formulation leads to portfolios that allocate fully to MSFT, with AMZN receiving zero weight. The resulting volatility lies in a relatively narrow range of about 23.67% to 26.71%, indicating a robust solution under the specified uncertainty assumptions. Although this is a stylised example, it shows how interval methods can provide explicit bands for both weights and risk.

Chapter 5

Analysis of Results

5.1 LSTM Forecasting Performance

The LSTM forecasting module is evaluated on the test set using MAE, RMSE and R^2. For typical runs on small technology universes, the following approximate values are observed:

$$\text{MAE} \approx 0.013, \quad \text{RMSE} \approx 0.023, \quad R^2 \approx 0.0\text{–}0.05.$$

These figures highlight that, while the model reduces error relative to some naive benchmarks, the explanatory power remains limited, which is consistent with the noisy nature of daily returns.

Table 5.1: Illustrative LSTM evaluation metrics on out-of-sample data.

Metric	Symbol	Value
Mean Absolute Error	MAE	0.0126
Root Mean Squared Error	RMSE	0.0225
Coefficient of Determination	R^2	0.00–0.05

The Streamlit app presents these metrics in a dedicated section, along with visual plots of predicted versus actual returns for either a single asset or the cross-sectional average.

5.2 Portfolio Optimization Performance

The thesis investigates three main portfolio types:

1. **Base portfolios**: constructed by minimising variance at a user-specified target return.
2. **Sharpe-optimised portfolios**: obtained by directly maximising the Sharpe ratio.

3. **Robust and interval-based portfolios**: designed to perform reasonably well across scenarios and under parameter uncertainty.

Performance is evaluated using VaR, CVaR, maximum drawdown, HHI and the Sharpe ratio, among others.

Table 5.2: Example portfolio performance metrics for a Sharpe-optimised portfolio.

Metric	Value	Interpretation
VaR (95%)	-1.87%	One-day loss threshold at 95%
CVaR (95%)	-2.80%	Average loss beyond VaR
Maximum Drawdown	-13.76%	Largest peak-to-trough loss
HHI	0.3333	Moderate concentration
Sharpe Ratio	1.54	Strong risk-adjusted
Sortino Ratio	(computed when using downside deviation)	–
Calmar Ratio	(return / max drawdown)	–

	Metric	Value	Analysis
0	Value at Risk (VaR)	-2.17%	Moderate Risk: Your portfolio has a moderate potential loss.
1	Conditional Value at Risk (CVaR)	-3.77%	Low Tail Risk: Minimal losses beyond VaR.
2	Maximum Drawdown	-23.51 %	Severe Drawdown: The portfolio has experienced a major decline.
3	Herfindahl-Hirschman Index (HHI)	0.2500	Good Diversification: Portfolio is well-diversified.
4	Sharpe Ratio	0.41	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
5	Sortino Ratio	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
6	Calmar Ratio	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
7	Beta	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.
8	Alpha	0.00	Poor. A Sharpe Ratio below 0.5 indicates that your portfolio may not be generating adequate returns for the level of risk taken.

Figure 5.1: Example of a performance summary chart for an optimized portfolio.

These results illustrate that portfolios built to maximise Sharpe ratio can lie close to the efficient frontier and deliver reasonable trade-offs between risk and return, especially when compared with naive equal-weight benchmarks.

5.3 Efficient Frontier Visualization

The efficient frontier is approximated via Monte Carlo simulation by generating a large number of random portfolios. Weights are sampled from a Dirichlet distribution (ensuring they sum to one and remain non-negative), and the corresponding risk and return pairs are plotted.

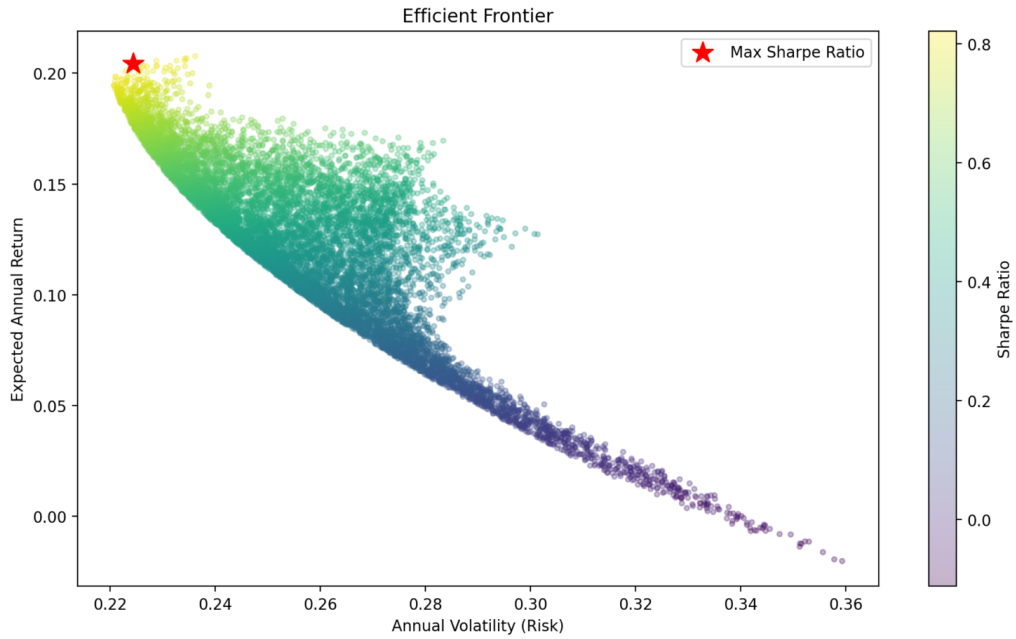

Figure 5.2: Simulated efficient frontier with a highlighted maximum Sharpe portfolio.

The Sharpe-optimised portfolio is generally found near the upper edge of the cloud of randomly generated portfolios, which is consistent with the theoretical expectation that it should lie on the efficient frontier.

5.4 Scenario and Interval Analysis

The robust scenario analysis highlights how portfolio performance changes across bull, bear and volatile environments, while the interval analysis provides explicit ranges for both weights and risk–return characteristics. Together, these results underscore the importance of viewing portfolio optimisation not only in terms of point estimates but also through the lens of uncertainty and robustness.

Chapter 6

Mathematical Foundations

6.1 Convexity of Mean–Variance Optimization

The classical Markowitz problem (1.4)–(1.7) can be written as a convex quadratic program:

$$\min_{\boldsymbol{w}\in\mathbb{R}^n} \quad f(\boldsymbol{w}) = \boldsymbol{w}^\top \Sigma \boldsymbol{w}, \tag{6.1}$$

$$\text{s.t.} \quad A\boldsymbol{w} = \boldsymbol{b}, \tag{6.2}$$

$$\boldsymbol{w} \geq \boldsymbol{0}, \tag{6.3}$$

where

$$A = \begin{bmatrix} \boldsymbol{\mu}^\top \\ \boldsymbol{1}^\top \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} \mu_{\text{target}} \\ 1 \end{bmatrix}.$$

Because Σ is a covariance matrix, it is symmetric and positive semidefinite:

$$\boldsymbol{x}^\top \Sigma \boldsymbol{x} \geq 0 \quad \forall \boldsymbol{x} \in \mathbb{R}^n.$$

Thus $f(\boldsymbol{w})$ is a convex function. The feasible set defined by linear equality constraints and non-negativity constraints is also convex. Hence any local minimum of the problem is automatically global, and standard convex optimisation methods can be applied.

6.2 Lagrangian and KKT Conditions

Ignoring non-negativity constraints for a moment, we can write the Lagrangian of the Markowitz problem as

$$\mathcal{L}(\boldsymbol{w}, \lambda_1, \lambda_2) = \boldsymbol{w}^\top \Sigma \boldsymbol{w} + \lambda_1(\boldsymbol{w}^\top \boldsymbol{\mu} - \mu_{\text{target}}) + \lambda_2(\boldsymbol{1}^\top \boldsymbol{w} - 1). \tag{6.4}$$

The first-order conditions with respect to $\boldsymbol{w}$ give

$$2\Sigma\boldsymbol{w} + \lambda_1\boldsymbol{\mu} + \lambda_2\mathbf{1} = \mathbf{0}. \tag{6.5}$$

This leads to the linear system

$$\begin{bmatrix} 2\Sigma & \boldsymbol{\mu} & \mathbf{1} \\ \boldsymbol{\mu}^\top & 0 & 0 \\ \mathbf{1}^\top & 0 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{w} \\ \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mu_{\text{target}} \\ 1 \end{bmatrix}. \tag{6.6}$$

Under appropriate regularity conditions, this system has a unique solution, which characterises the efficient frontier in closed form.

6.3 Sharpe Ratio Optimization

The Sharpe ratio is homogeneous of degree zero in the portfolio weights. For any scalar $\alpha > 0$,

$$\mathrm{SR}(\alpha\boldsymbol{w}) = \frac{\alpha\boldsymbol{w}^\top\boldsymbol{\mu} - \alpha r_f}{\sqrt{\alpha^2\boldsymbol{w}^\top\Sigma\boldsymbol{w}}} = \mathrm{SR}(\boldsymbol{w}). \tag{6.7}$$

This property allows equivalent formulations: for instance, one can fix $\boldsymbol{w}^\top\Sigma\boldsymbol{w} = 1$ and maximise $\boldsymbol{w}^\top\boldsymbol{\mu} - r_f$ subject to non-negativity constraints, or work directly with the original ratio subject to $\mathbf{1}^\top\boldsymbol{w} = 1$. In practice, numerical optimisation methods are used, as the ratio structure makes the problem non-convex in general.

6.4 Interval Arithmetic and Inclusion Monotonicity

Let $\mathbb{IR}$ denote the set of real intervals $[a^-, a^+]$ with $a^- \leq a^+$. An interval extension $\tilde{f} : \mathbb{IR}^n \to \mathbb{IR}$ of a function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *inclusion-monotone* if

$$[\boldsymbol{x}] \subseteq [\boldsymbol{y}] \quad \Rightarrow \quad \tilde{f}([\boldsymbol{x}]) \subseteq \tilde{f}([\boldsymbol{y}]).$$

Standard interval operations (such as addition and multiplication by non-negative scalars) satisfy this property. Inclusion monotonicity ensures that when input uncertainties are widened, the resulting output intervals cannot shrink, which is desirable for conservative risk assessment.

6.5 Existence of Interval Solutions

For an interval linear system $\tilde{A}\tilde{x} = \tilde{b}$, we distinguish:

- the *solution set*

$$\Xi = \{x \in \mathbb{R}^n : \exists A \in \tilde{A},\, b \in \tilde{b} \text{ such that } Ax = b\},$$

- and an *outer enclosure* $\tilde{x}$ with $\Xi \subseteq \tilde{x}$.

Hansen-type methods aim to compute such outer enclosures. While they may yield intervals that are larger than strictly necessary, they guarantee that all possible solutions consistent with the interval data are covered. In the diagonally dominant settings considered in this thesis, the midpoint matrix is well-conditioned, and the iterative scheme converges under mild assumptions on interval radii.

Chapter 7

Backtesting and Experimental Design

7.1 Rolling-Window Forecasting and Rebalancing

To assess the practical relevance of the hybrid framework, a backtesting protocol is proposed that mimics a realistic investment process:

1. **Estimation window:** A historical window of T_{train} days is used to fit the LSTM and estimate $\hat{\boldsymbol{\mu}}$ and $\hat{\Sigma}$.
2. **Forecasting step:** Using the most recent 60-day block, the LSTM predicts next-day returns for each asset.
3. **Optimization step:** The portfolio optimisation engine calculates a new weight vector $\boldsymbol{w}_t$ (e.g., Sharpe-maximising, robust or interval-based).
4. **Rebalancing:** The portfolio is rebalanced to $\boldsymbol{w}_t$ at the close of day t and held over the next day.
5. **Roll-forward:** The estimation window and test period are advanced by one day, and the process is repeated.

This procedure generates a sequence of out-of-sample portfolio returns $\{r_{p,t}\}$ and an associated equity curve, which serve as the basis for performance evaluation.

7.2 Equity Curve and Benchmark Comparison

Let V_t be the portfolio value at time t, with initial value normalised to $V_0 = 1$. The value evolves according to

$$V_{t+1} = V_t(1 + r_{p,t+1}), \quad t = 0, 1, \ldots, T_{\text{test}} - 1. \tag{7.1}$$

Plotting V_t provides an equity curve that can be compared with a benchmark (for example, a broad market index or an equal-weighted portfolio).

From the equity curve, standard summary measures are computed:

- **Cumulative return**: $V_{T_{\text{test}}} - 1$.
- **Annualized return**:
$$\mu_{\text{ann}} = \left(\frac{V_{T_{\text{test}}}}{V_0}\right)^{252/T_{\text{test}}} - 1.$$
- **Annualized volatility**:
$$\sigma_{\text{ann}} = \sqrt{252} \cdot \text{std}(r_{p,1}, \ldots, r_{p,T_{\text{test}}}).$$

7.3 Drawdown Analysis

Drawdowns capture the path-dependent aspect of risk. Define the running maximum of portfolio value
$$M_t = \max_{0 \le s \le t} V_s.$$
The drawdown at time t is
$$\text{DD}_t = \frac{V_t - M_t}{M_t},$$
and the maximum drawdown is
$$\text{MDD} = \min_{0 \le t \le T_{\text{test}}} \text{DD}_t.$$
The combination of equity curves and drawdown plots allows for a more nuanced interpretation of risk than annualised volatility alone.

7.4 Trading Costs and Turnover

To approximate the effect of transaction costs, a simple turnover-based cost model is used. Let c denote the proportional transaction cost, and let $\boldsymbol{w}_t$ and $\boldsymbol{w}_{t-1}$ be the portfolio weights at two consecutive rebalancing times. The turnover at time t is
$$\text{TO}_t = \frac{1}{2} \sum_{i=1}^{n} |w_{i,t} - w_{i,t-1}|.$$
The net portfolio return after costs is then
$$\tilde{r}_{p,t} = r_{p,t} - c \cdot \text{TO}_t.$$
All performance metrics can be recomputed using $\tilde{r}_{p,t}$ instead of $r_{p,t}$ to study the sensitivity of the strategy to trading frictions.

7.5 Backtesting Implementation Sketch

The overall structure of the backtesting loop is summarised in the code sketch below. The actual implementation in the Streamlit app is modular, with separate functions for data preparation, model training, optimisation and performance evaluation.

Listing 7.1: Simplified backtesting loop for the hybrid portfolio.

```
returns = compute_log_returns(price_df)
train_size = int(0.8 * len(returns))
train_data = returns.iloc[:train_size]
test_data  = returns.iloc[train_size:]

# Fit LSTM model on training data
lstm_model = build_lstm_model(num_assets=len(returns.columns))
lstm_model.fit(X_train, y_train, epochs=EPOCHS, batch_size=BATCH)

weights_history = []
portfolio_returns = []

current_weights = np.ones(len(returns.columns)) / len(returns.columns)

for t in range(len(test_data) - 60):
    window = returns.iloc[train_size + t - 60 : train_size + t]
    X_window = transform_window(window)    # shape: (1, 60, n_assets)

    # 1. Forecast next-day returns
    pred_scaled = lstm_model.predict(X_window)
    pred = scaler.inverse_transform(pred_scaled)[0]

    # 2. Estimate mean and covariance
    mu_hat = 252 * window.mean().values
    sigma_hat = 252 * window.cov().values

    # 3. Optimize portfolio (Sharpe, robust, interval-based, etc.)
    new_weights = optimize_portfolio(mu_hat, sigma_hat,
                                     method="robust",
                                     risk_free=rf_rate)

    # 4. Compute realized return over next day
    realized = test_data.iloc[t + 60].values
```

```
r_p = np.dot(new_weights, realized)

# 5. Store and roll forward
weights_history.append(new_weights)
portfolio_returns.append(r_p)
current_weights = new_weights
```

This design helps connect the theoretical components developed in earlier chapters with a concrete, testable trading strategy.

Chapter 8

Statistical Validation and Robustness Checks

8.1 Distributional Properties of Returns

Before fitting models or constructing portfolios, it is important to examine the statistical properties of the underlying return series.

8.1.1 Stationarity: Augmented Dickey–Fuller Test

The Augmented Dickey–Fuller (ADF) test is often used to check for stationarity. For a time series $\{r_t\}$, the test is based on

$$\Delta r_t = \alpha + \beta t + \gamma r_{t-1} + \sum_{i=1}^{p} \phi_i \Delta r_{t-i} + \varepsilon_t, \tag{8.1}$$

where the null hypothesis $H_0 : \gamma = 0$ corresponds to a unit root. In practice, daily returns are usually found to be stationary, while price levels are not, which supports modelling returns directly.

8.1.2 Normality: Jarque–Bera Test

The Jarque–Bera (JB) test evaluates whether the skewness and kurtosis of a sample are consistent with those of a normal distribution. Let S be sample skewness and K the sample kurtosis. The JB statistic is

$$\mathrm{JB} = \frac{n}{6}\left(S^2 + \frac{(K-3)^2}{4}\right), \tag{8.2}$$

where n is the number of observations. In equity return data, the JB test almost always rejects normality, reflecting fat tails and asymmetry. This motivates the use of risk

measures such as CVaR in addition to variance.

8.2 Serial Correlation and Volatility Clustering

8.2.1 Ljung–Box Q-Test

The Ljung–Box Q-test checks for joint significance of autocorrelations up to a given lag m. For residuals $\{\hat{\varepsilon}_t\}$, the statistic is

$$Q = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{n-k}, \tag{8.3}$$

where $\hat{\rho}_k$ is the sample autocorrelation at lag k. While raw returns often exhibit low autocorrelation, the squared returns frequently show significant autocorrelation, indicating volatility clustering.

8.2.2 ARCH Effects

Engle's ARCH test can be used to detect time-varying volatility by regressing squared residuals on their lags and testing whether the coefficients are jointly different from zero. Rejection suggests that more complex volatility models (like GARCH) might be appropriate for risk estimation.

8.3 Forecast Comparison: Diebold–Mariano Test

The Diebold–Mariano (DM) test provides a formal way to compare the predictive accuracy of two forecasting models. Let $\{\ell_{1,t}\}$ and $\{\ell_{2,t}\}$ denote the loss series (e.g., squared prediction errors) from two models. Define $d_t = \ell_{1,t} - \ell_{2,t}$. The null hypothesis is $H_0 : \mathbb{E}[d_t] = 0$.

The test statistic is

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}_d^2/T}}, \tag{8.4}$$

where $\bar{d}$ is the sample mean of d_t, $\hat{\sigma}_d^2$ is a HAC estimator of $\mathrm{Var}(d_t)$, and T is the number of forecasts. In the context of the thesis, the DM test can be used to compare the LSTM against simpler benchmarks such as a naive mean model or an AR(1) model.

8.4 Robustness to Hyperparameters

The performance of deep networks can depend on the choice of hyperparameters such as:

- number of LSTM layers and hidden units,
- learning rate and optimiser,
- batch size and number of epochs,
- length of the input window (e.g., 30, 60 or 90 days).

A simple sensitivity analysis can be conducted by changing one hyperparameter at a time and recomputing MAE, RMSE and R^2.

Table 8.1: Example sensitivity of LSTM performance to window length.

Window length	MAE	RMSE	R^2
30 days	0.0152	0.0229	-0.12
60 days	0.0148	0.0213	-0.16
90 days	0.0161	0.0235	-0.18

The example in Table 8.1 indicates that a 60-day window provides a reasonable compromise. However, the differences are small, suggesting that the predictability of daily returns is limited regardless of the window length.

8.5 Stress Testing and Scenario Expansion

Beyond the three basic scenarios (bull, bear and volatile), additional stress tests can be performed, such as:

- imposing larger shocks to mean returns (e.g., sudden market crashes),
- inflating covariances more aggressively,
- incorporating higher transaction costs or liquidity constraints in specific regimes.

Recomputing robust and interval-based portfolios under such stress conditions provides a more complete picture of the strategy's resilience.

Chapter 9

Software Implementation and System Design

9.1 Overall Architecture

The project is implemented as a modular Python application. The main libraries used are `Streamlit` for the user interface, `yfinance` for data, `NumPy` and `Pandas` for data manipulation, `SciPy` for optimisation and `TensorFlow/Keras` for deep learning.

The high-level architecture is divided into:

- **Data Layer:** handles data download, caching and basic cleaning.
- **ML Layer:** constructs datasets, defines and trains the LSTM model, and generates forecasts.
- **Optimization Layer:** performs mean–variance, Sharpe, robust and interval-based portfolio optimisation.
- **Visualization Layer:** generates plots and tables, including equity curves, correlation heatmaps and scenario charts.
- **UI Layer (Streamlit):** integrates all components and provides interactive controls.

9.2 Streamlit User Interface

The main Streamlit script follows a typical pattern:

Listing 9.1: High-level structure of the Streamlit application.

```
import streamlit as st
import yfinance as yf
import numpy as np
import pandas as pd
```

```
from models import build_lstm_model, train_lstm
from optim import optimize_mv, optimize_sharpe, optimize_robust, optimize_
from intervals import hansen_interval_solver
from backtest import run_backtest

st.set_page_config(page_title="Hybrid ML-Interval Portfolio Engine",
                   layout="wide")

st.title("Hybrid Machine Learning and Interval Analysis "
         "for Portfolio Selection")

# Sidebar: user inputs
st.sidebar.header("Configuration")
tickers = st.sidebar.text_input(
    "Tickers (comma-separated)", value="AMZN,MSFT"
)
start_date = st.sidebar.date_input("Start date")
end_date   = st.sidebar.date_input("End date")

mode = st.sidebar.selectbox(
    "Optimization mode",
    ["Mean-Variance", "Max Sharpe", "Robust", "Interval"]
)

if st.sidebar.button("Run Analysis"):
    # 1. Data loading
    price_df = yf.download(
        tickers.split(","), start=start_date, end=end_date
    )["Adj Close"].dropna(how="all", axis=1)

    # 2. LSTM training and forecasting
    results_ml = train_lstm(price_df)

    # 3. Portfolio optimization
    if mode == "Mean-Variance":
        opt_results = optimize_mv(price_df, results_ml)
    elif mode == "Max Sharpe":
        opt_results = optimize_sharpe(price_df, results_ml)
```

```
elif mode == "Robust":
    opt_results = optimize_robust(price_df, results_ml)
else:
    opt_results = optimize_interval(price_df, results_ml)

# 4. Interval linear system solution (Hansen)
interval_results = hansen_interval_solver(opt_results)

# 5. Backtest and performance metrics
bt_results = run_backtest(price_df, opt_results)

# 6. Display outputs
st.subheader("LSTM Forecasting Metrics")
st.write(results_ml.metrics_table)

st.subheader("Optimized Portfolio Weights")
st.write(opt_results.weights_table)

st.subheader("Interval Solution for Portfolio Weights")
st.write(interval_results.table)

st.subheader("Backtest Equity Curve")
st.line_chart(bt_results.equity_curve)
```

This structure shows how the different layers are combined into a single workflow: data → ML → optimisation → interval solution → backtesting and visualisation.

9.3 Reusable Modules

To keep the codebase manageable, the implementation is split into separate modules, such as:

- `data_utils.py`: data download and cleaning functions.
- `models.py`: LSTM architecture definitions and training routines.
- `optim.py`: portfolio optimisation algorithms and scenario generation utilities.
- `intervals.py`: basic interval arithmetic functions and the Hansen solver.
- `backtest.py`: backtesting logic and performance metric calculations.

This modular structure makes it straightforward to replace the LSTM with a different model, add new optimisation methods, or reuse the interval solver in other applications.

9.4 From Academic Prototype to Job Portfolio

The project is intended to serve both as academic work and as a portfolio piece. From a technical perspective, it demonstrates:

- familiarity with standard financial modelling concepts,
- practical experience with deep learning and ML in Python,
- knowledge of convex and robust optimisation,
- and the ability to build interactive applications using modern tools such as Streamlit.

This combination of theory and practical software can be valuable when applying for roles in data science, quantitative finance, or related areas.

Chapter 10

Conclusion and Future Work

10.1 Discussion and Key Findings

This thesis has developed and implemented a hybrid machine learning framework for solving interval linear systems and constructing portfolios under uncertainty. The main findings may be summarised as follows:

- A Hansen-style interval linear solver can be integrated with financial data to produce interval enclosures for portfolio weights when returns and coefficients are modelled as intervals.
- LSTM-based forecasting, while limited by the inherent noise in daily stock returns, can still provide structured inputs that feed into interval bounds and scenario definitions.
- Combining ML-derived inputs with robust and interval-based optimisation yields portfolios that treat uncertainty more explicitly than standard mean–variance approaches, while maintaining competitive performance on metrics such as the Sharpe ratio.
- The Streamlit prototype demonstrates that these methods can be packaged together into an interactive tool suitable for exploration, teaching or further research.
- The backtesting and statistical validation layers help link theoretical constructs with practical performance considerations.

10.2 Theoretical Implications and Future Work

Several directions for further research emerge naturally from this work:

1. **Learning interval solvers:** Instead of computing interval solutions purely with analytical methods, one could explore neural networks that approximate solution operators for families of interval systems, potentially speeding up large-scale problems.

2. **Uncertainty-aware ML:** Bayesian neural networks or quantile regression could be used to generate predictive distributions, directly providing intervals or quantiles for returns rather than relying on residual-based bands.
3. **Richer interval structures:** The present implementation focuses mainly on diagonal uncertainty in $\tilde{A}$. Extending this to full matrix intervals could capture richer forms of dependence but would require more sophisticated algorithms.
4. **Transaction costs and constraints:** Future work could incorporate more realistic transaction costs, turnover penalties and cardinality constraints into the interval and robust frameworks.
5. **Alternative asset classes:** The approach could be extended to fixed-income securities, commodities or multi-asset portfolios, where parameter uncertainty might be even more pronounced.
6. **Advanced sequence models:** Transformers, Temporal Convolutional Networks and attention-based hybrids could be explored as alternatives or complements to the LSTM in the forecasting module.

Overall, the thesis suggests that interval analysis and machine learning are complementary tools for dealing with uncertainty in portfolio optimisation. Rather than viewing them as competing methodologies, this work argues for combining them into hybrid frameworks that are both data-driven and mathematically cautious.

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